

# Disaster Blackbox & Smart Surveillance System

A Groundbreaking Life-Saving Technology for Earthquake, Fire Hazard, and Intrusion Response

## 1. GENERAL / ECONOMY MODEL (ECO-SHIELD EDITION)

Target Customers: General Residences, Small Shops, Hostels/Mess, and Middle-Class Families

Highly affordable and easily mountable on walls or floors.

### Sensors Used & Their Functionalities

- **MPU6050 Accelerometer (Earthquake Detector):**
  - **Function:** Detects initial earthquake vibrations (P-Wave).
  - **Why it's the best:** Detects danger several seconds before the main high-intensity waves hit, activating the alarm in advance.
- **MQ-2 Gas & Smoke Sensor:**
  - **Function:** Detects gas leaks and initial fire smoke.
  - **Why it's the best:** Sends immediate warnings as soon as toxic or cooking gas disperses into the air, preventing major accidents.
- **PIR Motion Sensor (Security & Tracking):**
  - **Function:** Detects human movement and human presence on the floor.
  - **Why it's the best:** Functions as a security alarm against intruders/burglars at night and helps identify trapped individuals on the floor during disasters.
- **SIM800L GSM Module:**
  - **Function:** Sends calls and text messages via the mobile network.
  - **Why it's the best:** In the event of a disaster or break-in, it automatically dials the Police and Fire Service, broadcasting the precise location using a pre-recorded voice message.

### System Workflow (How it Works)

**Step 1 (Disaster Detection):** Upon sensing an earthquake or fire, the device triggers a loud siren to alert everyone inside.

**Step 2 (Automated Protection):** Via a relay module, it automatically shuts off the main electrical and gas lines to prevent electrical fires from spreading.

**Step 3 (Emergency Calling):** Independent of the internet, it direct-calls Emergency Services via the SIM card and plays an audio broadcast:

"Emergency help needed! Fire reported at Road No. 4, Sector 10, Uttara."

**Step 4 (Guided Rescue):** When rescuers arrive, floor-level beacon lights illuminate, and the unit plays calming guidance instructions via loudspeakers to trapped occupants.

### Production Cost & Financial Analysis

SINGLE UNIT COST

**BDT 2,400**

Estimated prototype cost

MASS PRODUCTION (1M)

**BDT 1,450 / pc**

Chip costs drop with bulk import

MARKET POTENTIAL

**Retail < BDT 5,000**

Up to ~300% profit margin

## 2. PRO / GOVERNMENT MODEL (TITAN-GUARD EDITION)

Target Customers: Government Offices, Banks, High-rise Commercial Buildings, Hospitals, & Garment Factories

Military-grade protection equipped with fully autonomous technology.

### Sensors Used & Their Functionalities

- **ADXL355 Industrial Seismic Sensor:**
  - **Function:** Detects micro-vibrations and structural deviations in buildings.
  - **Why it's the best:** Accurately differentiates ambient noise/vibrations from true seismic activity, eliminating false alarms.
- **mmWave Radar (LD2410):**
  - **Function:** Detects heartbeats and respiration rates of stationary and moving individuals.
  - **Why it's the best:** Where standard cameras fail in dense smoke or total darkness, this radar accurately counts and locates living or unconscious individuals on a floor.
- **AMG8833 Thermal Grid Sensor:**
  - **Function:** Reads human thermal signatures across an 8x8 grid.
  - **Why it's the best:** Maps precise human coordinates without camera optics, guaranteeing absolute privacy.
- **LoRa Mesh Networking & Satellite Backup:**
  - **Function:** Transmits data within a 2–5 km radius without cellular or internet infrastructure.
  - **Why it's the best:** Transmits critical data from underneath collapsed building rubble even if cell towers fail.

### System Workflow (How it Works)

**Step 1 (Autonomous Action):** Instantly upon detection, it activates building sprinkler systems and parks/locks elevator doors at safe exit floors.

**Step 2 (Short-Circuit Mapping):** Pinpoints the exact floor and junction of short circuits, rendering a dynamic layout in the main control room.

**Step 3 (Live Autonomous Call & AI Telemetry):** Transmits real-time AI-processed data directly to emergency services:

*"Short circuit fire on 5th floor. 3 individuals trapped."*

**Step 4 (Under-Rubble Blackbox Survivability):** Built with a ceramic-insulated, waterproof armor casing that survives structural collapse. Its internal LiFePO4 battery transmits continuous radio signals for 7 days:

*"We are here. 2 survivors located in the far-east corner."*

### Production Cost & Financial Analysis

#### SINGLE UNIT COST

**BDT 15,500**

Estimated prototype cost

#### MASS PRODUCTION (1M)

**BDT 8,200 / pc**

Custom PCB & scaled R&D amortize cost

#### MARKET POTENTIAL

**BDT 30,000 – 35,000**

Up to ~400% profit margin

### 3. WHY OUR PROJECT IS UNIQUE & SUPERIOR (USPS)

**Life Preservation as Highest Priority:** Beyond sounding alarms, it reports exact victim counts and coordinates from deep inside debris, dramatically accelerating rescue operations and saving lives.

**Dual Protection System:** Offers disaster protection while simultaneously securing multi-million dollar state assets and private properties against burglary and unauthorized intrusion.

**Offline & Grid-Independent Reliability:** Fully operational even during complete power failures or cellular grid downtime due to long-life internal batteries and independent mesh radio connectivity.

**Affordable & Scalable Customization:** Tailored into two distinct tiers to accommodate all economic demographics across the country—a key driver for massive market adoption and commercial success.